



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Network Design for Faculty of Computing Block N28B

(by 404 NOT FOUND)

FACULTY OF COMPUTING
SEMESTER 1 SESSION 2024/2025

SECR1213 - NETWORK COMMUNICATIONS

PROJECT TASK 6: DOCUMENTATION AND REFLECTION

TASK 6A: MAKING A GROUP REPORT

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SECTION: 10

Abstract

In this project, our group has been tasked to design the network plan for the Faculty of Computing's new block, N28B. The budget that has been given to us is 3 million ringgit. In task 1, we designed the floor plans of the two-storey building that includes four new labs, a video conferencing room for virtual meetings, a hybrid classroom designed for flexible learning, and a student lounge with WIFI connectivity for work and relaxation purposes. In task 2, we conducted an interview for preliminary analysis and a feasibility assessment which covers economic, technical, and operational feasibility. In task 3, we did research on suitable devices for our project and compared between different specifications to find the best one. We then calculated the cost needed to acquire all devices. In task 4, we identified the work areas of each floor from their floor plan. Besides, we planned the network diagram for each room, and determined the cable distribution of the two floors. We then calculated the number of switch ports, patch cords, RJ45 connectors, and the cable length. In task 5, we did subnetting and assigned IP addresses to each lab and room.

Table of Contents

1.0 Introduction.....	4
2.0 Project background and an overview of the client's current status and issue.....	5
3.0 A compiled solution of Task 1-5.....	6
3.1 Task 1 - Project Setup - Floor Plan.....	6
3.1.1 Ground Floor.....	6
3.1.2 First Floor.....	7
3.2 Task 2 - Initial Design - Preliminary Analysis.....	8
3.2.1 Questions.....	8
3.2.2 Feasibility Assessment.....	12
3.3 Task 3 - Choosing the Appropriate LAN Devices.....	15
3.3.1 List of devices.....	15
3.3.2 Expected cost.....	36
3.3.3 Reflection.....	37
3.4 Task 4 - Making The Connections - LAN and WAN.....	39
3.4.1 Work areas on the floor plan.....	39
3.4.2 Network Diagram.....	41
3.4.3 Cables & Connections.....	46
3.5 Task 5 - IP Addressing Scheme.....	51
3.5.1 Subnet Mask.....	51
3.5.2 Range of IP address, network and broadcast address for the subnet.....	52
3.5.3 Range of IP address for user types based on the area.....	53
4.0 Conclusion.....	56
5.0 Team Members and Responsibilities.....	57
6.0 References.....	58
Appendices.....	59
1. First Meeting Minutes.....	59
2. Second Meeting Minutes.....	63
3. Third Meeting Minutes.....	65
4. Fourth Meeting Minutes.....	68
5. Fifth Meeting Minutes.....	72
6. Pictures.....	75
7. Financial Budget.....	76

Table of Figures

Figure 1: Ground Floor Floor Plan.....	6
Figure 2: First Floor Floor Plan.....	7
Figure 3: Work Area on the Ground Floor.....	39
Figure 5: Lecturer Hall Network Diagram.....	41
Figure 6: General Lab Network Diagram.....	41
Figure 7: Student Lounge Network Diagram.....	42
Figure 8: Cafe Network Diagram.....	42
Figure 9: Video Conferencing Room Network Diagram.....	43
Figure 10: Hybrid Classroom Network Diagram.....	43
Figure 11: Rest Area Network Diagram.....	43
Figure 12: Server Room Network Diagram.....	43
Figure 13: Office Network Diagram.....	44
Figure 14: Cisco Network Lab Network Diagram.....	44
Figure 15: Embedded Lab Network Diagram.....	45
Figure 16: Maintenance Room Network Diagram.....	45
Figure 17: Cable distribution of ground floor.....	46
Figure 18: Cable distribution of the first floor.....	46
Figure 19: Ground Floor Plan with label.....	55
Figure 20: First Floor Plan with label.....	55

1.0 Introduction

The project was aimed at helping the Faculty of Computing to build a Local Area Network (LAN) for the new double-storey building. The new building consisted of 4 new labs, a video conferencing room for virtual project meetings, a hybrid classroom and a student lounge for giving the students work and relaxation. For the labs, there are two general purpose labs, 1 Cisco Network lab and 1 embedded lab. Each lab will provide a comfortable space for both working and learning space for 30 workstations inside every lab.

The project's scope is to create a system that is easy to maintain while still having a scalable network for future expansion and wireless access. The system should actually be able to enhance overall performance in addition to addressing the expansion of the Faculty of Computing (FC) and safeguarding the system from network intrusions such as denial-of-service assaults, Internet Worms and e-business application attacks. Through access control lists (ACLs), the system may additionally handle aspects like Quality of Service and hardware security, as well as the high performance of the main backbone. However, the suggested solution will allow for a secure VPN connection from distant areas, creating a more fulfilling environment for faculty members and students. This project will also involve designing the new building's floor plan and implementing the network devices needed for the new building by researching the design process and using case studies. In fact, in order to make sure that every network device in the new building can work, the length of the wired connection—the connection between the devices and the network—will also be assessed.

The objective of the project is to give the Faculty of Computing improved equipment and an environment that supports its expansion while also advancing with cutting-edge technology without undergoing any further improvements for the following 20 years. The system will set up the lab using the fourth industrial revolution (4th IR) concept and guarantee that the network design will be both economical and appropriate for the size of this project.

2.0 Project background and an overview of the client's current status and issue

In this project, the client Faculty of Computing (FC) is preparing for growth in student and staff numbers, which are expected to increase by 15% over the next four years. To support this, FC plans to construct a new two –storey building comprising four labs, a video conferencing room, a hybrid classroom, and a student lounge. All labs will cater for a variety of needs, including general purpose labs, networking labs and embedded labs, while lounges are being constructed to provide students with work and recreational space and to allow them to access the Internet using Wi-Fi.

The client's current situation is that the existing system is outdated and experiencing difficulties with reliability, speed and security, and FC needs to be scalable to meet future needs. In addition, the existing network lacked the advanced features to support the 4IR (Fourth Industrial Revolution) initiative. Therefore, our goal is to create a reliable, efficient, and secure network. At the same time, ensure high-speed internet for teaching, research, and learning activities. This also includes installing dedicated equipment in all laboratories, supporting collaboration and virtual meetings in video conferencing rooms and hybrid classrooms, and providing seamless connectivity in student lounges.

There are several key considerations, one of which is the efficient use of space to balance the needs of learning, teaching and relaxation. Second, ensure cost-effectiveness while maintaining quality and functionality. Through careful planning and selection of appropriate equipment, the project aims to create a network that supports collaboration and innovation among students and staff. In addition, comfort and accessibility are prioritized to create an inclusive environment for all the users.

3.0 A compiled solution of Task 1-5

3.1 Task 1 - Project Setup - Floor Plan

3.1.1 Ground Floor

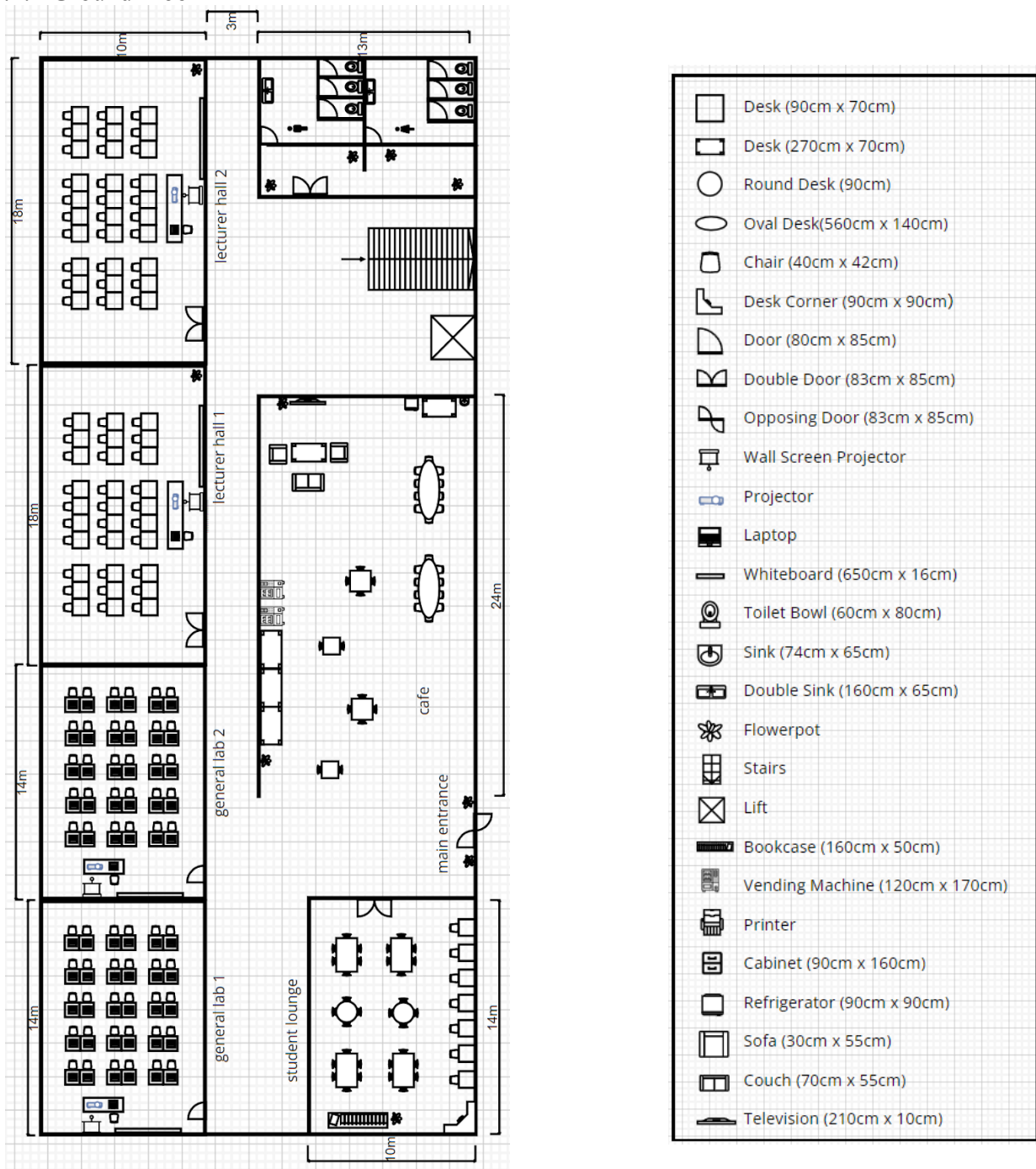


Figure 1: Ground Floor Floor Plan

3.1.2 First Floor

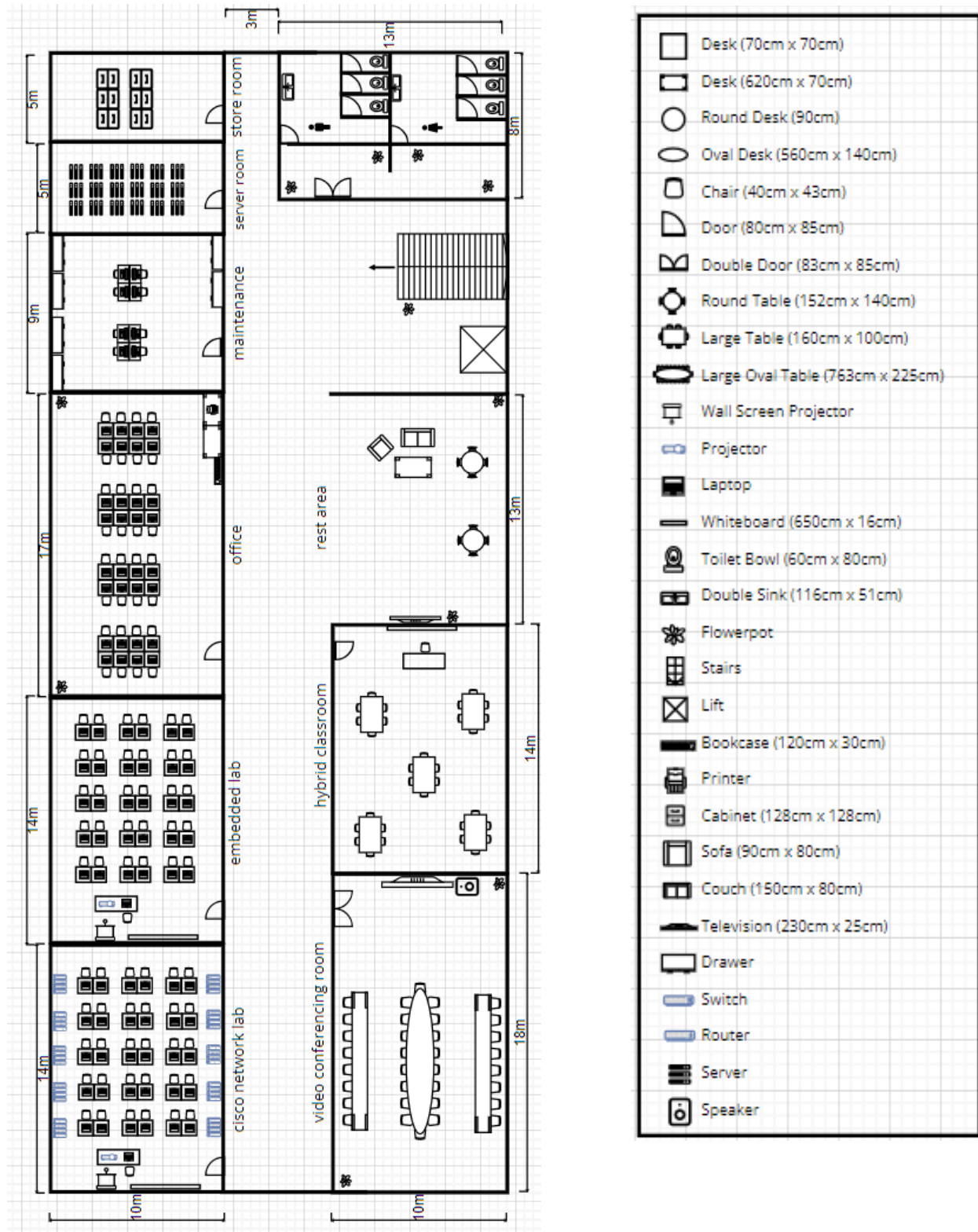


Figure 2: First Floor Floor Plan

3.2 Task 2 - Initial Design - Preliminary Analysis

3.2.1 Questions

1) What are the ideal computer specifications for the general-purpose labs, Cisco Network Lab, and Embedded Lab?

To support a variety of academic duties and demanding applications, high-performance computers are required for the labs, particularly the general-purpose labs, Cisco Network Lab, and Embedded Lab. Windows 11 and a 12th-generation Intel Core i7-12700K CPU, as well as an NVIDIA RTX 3090 graphics card, are among the suggested specifications. In order to manage several processes at once, these computers should also have 32GB of RAM. A 512GB SSD should be used for storage to enable quick read/write operations, which is particularly helpful for data-intensive applications like network configuration, simulations, and IoT development in the Embedded Lab. The workstations' ability to manage complex activities and upcoming technological demands is guaranteed by these specifications.

2) What are the required bandwidth and high-speed internet specifications for the labs, classrooms, and video conferencing rooms?

For the labs, classrooms, and video conferencing rooms to accommodate multiple users and bandwidth-demanding applications, dependable, fast internet is required. There are 30 workstations in each lab, and each one needs at least 10 Mbps, for a total of 300 Mbps per lab. Furthermore, additional bandwidth will be needed for servers and networked equipment in each lab, bringing the total bandwidth per lab to 500 Mbps. A total of 2 Gbps of bandwidth will be required for the four facilities. About 300 Mbps will be required by the video conferencing room in order to guarantee seamless virtual meetings. For a smooth experience, 300 Mbps will also be needed in the student lounge, where a large number of users will connect wirelessly. The building's total internet needs will be about 2.6 Gbps, which can be met by a 4G network at first. When 5G becomes available, it will be upgraded.

3) What are the security measures necessary for computers and the network to protect against threats?

Strong cybersecurity in the building is essential. To guard against dangers like malware, ransomware and phishing attempts, every workstation will be outfitted with firewalls, malware protection technologies, and anti-virus software. The practice will include regular security updates and system scans to guarantee that systems stay safe. Multiple layers of security, such as firewalls, intrusion detection and prevention systems (IDPS), and network segmentation, will be used to defend the network. Sensitive student and faculty data will be protected by these safeguards against malware outbreaks, denial-of-service attacks, and illegal access.

4) What measures should be taken to ensure physical security within the labs and building?

CCTV cameras will be positioned thoughtfully throughout the labs, hallways, and other crucial areas as part of the physical security measures to safeguard important equipment and data. Concave mirrors will be used to remove any blind spots. Smart card entry and other access control systems will be implemented at lab entrances and server rooms to ensure that only authorized staff can enter. A Zero Trust security model will be used in the server room, which houses vital network hardware and data storage, guaranteeing that only authorized personnel can enter. When executed consistently, this approach will prevent unauthorized access and theft of data or equipment.

5) What type of cabling and wiring infrastructure is best for labs and throughout the building to ensure high-speed internet connectivity that meets the needs of each lab and facility?

For the new Faculty of Computing building, a combination of cabling types was recommended to ensure high-speed connectivity and efficient operation. Therefore, the server room, video conferencing room, and Cisco Networking Lab will use fiber optic cables to enable long-distance data transmission and high-speed connections with minimal signal loss. Coaxial cables are an excellent choice for general labs and video

conferencing rooms because they can handle high definition video and fast data signals, which are important for presentations and interactive learning. They have low signal loss, so signals can be sent over long distances without losing quality. Additionally, low voltage control cables should be installed throughout all labs, the student lounge, and office areas to connect sensors, alarms, and control devices to effectively manage lighting and security systems to ensure a safe and secure environment for students. Fire alarm cables will also be installed throughout the building to connect essential equipment such as smoke detectors and alarm systems.

6) What is the benefit of having a dedicated server room in the faculty building?

A dedicated server room can enhance the network performance within the building by providing cooling and airflow to the servers. This will ensure that the equipment will run at peak efficiency without overheating. Besides, it offers a secure space where equipment can safely operate as it is equipped with security measures such as CCTV cameras and card entry systems. Having a centralized area for servers can also make activities such as monitoring, troubleshooting, and maintaining easier for staff since all the hardware is in the same place.

7) What steps should be taken to ensure optimal cooling and prevent overheating in high-performance server rooms and labs?

To ensure optimal cooling and prevent overheating in server room, and labs, it's important to follow several key strategies. First, enhance airflow by ensuring proper ventilation, providing enough space between hardware, and cleaning air filters regularly to prevent dust buildup. Next, place the hardware in a temperature-friendly environment, avoiding direct sunlight or heat sources. Invest in cooling solutions, such as air conditioners, cooling fans, or liquid cooling systems. Additionally, regularly removing dust from components, especially fans and radiators, and monitoring temperature levels with alerts can further prevent overheating and help maintain efficient operation.

8) How will the network ensure safe and efficient guest access to the student lounge while maintaining its isolation from the main network used by faculty, staff and students?

To ensure safe and efficient guest access to the student lounge while keeping faculty, staff, and students separate from the main network, a dedicated guest Wi-Fi network will be created. This guest network will have its own settings to prevent guests from accessing sensitive files or devices on the main network. Visitors can connect using a simple login page, which helps track who is using the web. The guest network will also have internet speed limits to ensure lecturers and students still have good access to their resources. By using strong security measures like WPA3 and regularly checking who is connecting, we can keep our network secure while allowing our guests to use the internet freely.

9) What is the suitable network topology to be used in the network of the building?

One of the suitable network topologies that can be used is the tree topology. It has a hierarchical structure which makes it support the grouping of different labs with its root and branch structure. It allows easy organization and management of security, permissions and data flow. Besides, it also supports high scalability since new devices can be easily added to the network with existing nodes. This allows the network to expand smoothly if the faculty continues to be upgraded in the future. When failure in the network occurs, it can easily be isolated and fixed because each branch is separated and does not affect the entire network.

10) How can network downtime be minimized in case of equipment failure or connectivity issues?

To minimize network downtime, the devices in the faculty should be monitored constantly using suitable tools and software such as OpManager, Paessler PRTG, and Auvik. These tools can track and measure our network's bandwidth, latency, throughput and more to alert us of any issues before major problems happen. Next, network redundancy and backup should be implemented for all of the network components within the faculty. This can be achieved by using multiple flows of traffic, like redundant switches and routers so that data can keep flowing in event that there is a failure.

3.2.2 Feasibility Assessment

1) Economic Feasibility

The project is financially feasible with a budget of RM 3 million, mostly for network systems and computer equipment. High-performance workstations, networking equipment, and specialized equipment for the Cisco Network Lab and Embedded Lab will be the main expenses. To guarantee seamless connectivity and engaging learning environments, the hybrid classroom and video conferencing room setup requires an investment in cutting-edge networking and audiovisual systems.

A sizeable amount of the financing will be allocated to building a dependable and secure network infrastructure in addition to computer hardware. With an emphasis on scalability for future growth, this comprises fiber-optic cable, network switches, routers, and Wi-Fi systems to enable high-speed internet connectivity throughout the facility. Access control, intrusion detection/prevention systems, firewalls, and other network security measures will also be crucial parts of the budget.

The project is financially feasible within the RM 3 million budget by concentrating on scalable, affordable solutions and carefully allocating expenditures to guarantee both immediate functionality and future-proofing.

2) Technical Feasibility

The expansion of the Faculty of Computing is highly technically feasible. The design incorporates advanced technologies such as fiber optic cabling in the server room and lab to ensure high-speed network connectivity and minimize data loss. These infrastructures are critical to support the wide range of academic activities, including high-performance computing tasks and video conferencing, that will take place in the new facility.

In addition to robust cabling, the lab will be equipped with powerful computers that meet the latest specifications for running demanding applications. High-end processors and ample memory will help tasks such as network simulation and IoT development run smoothly. In addition, the network architecture will include modern routers and switches to efficiently manage data traffic and ensure that all users have access to reliable Internet services.

Security is another key aspect of the technical feasibility assessment. The project plans to implement a variety of security measures such as firewalls, intrusion detection, and a separate guest Wi-Fi network to protect sensitive data. These policies will not only protect the main network, but also allow guests to access the Internet without compromising overall security. Overall, the technical component of the project was well suited to meet the needs of the Faculty of Computing and ensure a safe and efficient environment for students and staff.

3) Operational Feasibility

In terms of operations, the project is highly feasible. The implementations of facilities such as advanced labs and robust network systems ensures its reliability and scalability to users. With its centralized design, the network in the building can be managed and operated easily by the staff without causing disruptions if any issues happen. They are also able to monitor the systems frequently and troubleshoot them if there are equipment failures.

Besides, the high speed internet and cabling will provide seamless connectivity for users across the multiple labs in the building, especially those with high network requirements like the Cisco network lab and hybrid classroom. This allows daily teaching and learning activities to be carried out smoothly by both lecturers and students. Furthermore, both physical security and cybersecurity implementations in the building provides a safe environment to all sensitive data in the servers.

Overall, this project is shown to have high operational feasibility as it can be maintained by staff easily. It also has a reliable and efficient network system to support academic research in the faculty.

3.3 Task 3 - Choosing the Appropriate LAN Devices

3.3.1 List of devices

1. Switch

A switch is a device that connects multiple devices in a network, especially in a Local Area Network (LAN). It forwards data packets to and from those devices but only routes data to the specific devices it is intended for. When selecting a network switch, we need to consider features such as number of ports, speed of connections, and prices. For our network setup, we have chosen four suitable switches from the current market, which are the TP-Link TL-SG3452XP JetStream, Ubiquiti USW-Pro-48-PoE, and HPE Aruba 2930F JL262A. Each of the switches have their own advantages, with TP-Link TL-SG3452XP JetStream being the cheapest priced at RM4999 per unit. Next in line is the Ubiquiti USW-Pro-48-PoE priced at RM5779 per unit, and the most expensive one is the HPE Aruba 2930F JL262A priced at RM7509.99 per unit. In terms of specifications, all of them are similar with 48 PoE ports that are capable of transferring both power and data, while TP-Link TL-SG3452XP JetStream and Ubiquiti USW-Pro-48-PoE have a better switching capacity at 176 Gbps compared to HPE Aruba 2930F JL262A's 104 Gbps. After comparing their performance and price, we decided that the Ubiquiti USW-Pro-48-PoE is the most suitable for our building with its advanced Layer-3 switching protocols, powering capabilities, and higher throughput.

Product	TP-Link TL-SG3452XP JetStream	Ubiquiti USW-Pro-48-PoE	HPE Aruba 2930F JL262A
Specification	Ports: 48× 10/100/1000 Mbps RJ45 PoE+ Ports, 4× 10G SFP+ Slots	- Ports: 48×GbE RJ45 ports, 4×1/10G SFP+ ports - Switching	Ports: Ports: 48 x 10/100/1000 PoE+ ports, 4 x SFP ports

	• Switching capacity: 176 Gbps • Packet forwarding rate: 130.9 Mpps	capacity: 176 Gbps • Packet forwarding rate: 130.944 Mpps • Throughput: 88 Gbps	• Switching capacity: 104 Gbps • Throughput: Up to 77.4 Mpps
Price	RM4999	RM5779	RM7509.99
Reference	https://www.ect.my/tp-link-omada-sg3452xp-jetstream-48-port-gigabit-and-4-port-10ge-sfp-l2-managed-switch-with-48-port-poe	https://shopee.com.my/Ubiquiti-USW-Pro-48-PoE-10G-POE-Switch-48-ports-RJ45-Layer-3-1U-Rackmount-PoE-802.3bt-PoE--i.76579217.5515786036	https://www.sysnapse.com/ARUBA-2930F-48-Port-POE-Switch-with-4-SFP-Ports-JL262A?search=poe%20switch&sort=p.price&order=DESC

2. Router

A router is a key device in a network which connects multiple networks and directs data packets between them. It not only helps determine the path for data transmission but also ensures the efficiency and speed of network communication. For the Faculty of Computing's new building, the router must meet high performance requirements to handle the expected network load, support scalability for future growth, and include strong security features. The selection process included evaluating various routers based on Wi-Fi standards, Wi-Fi speeds, ports, and security features. We evaluated three routers, which are TP-Link Archer AX55 WL AX3000, Archer BE230 BE3600 Dual-Band, and TP Link Archer AX73 AX5400 Dual Band. First, Archer AX55 is priced at RM299 and offers dual-band Wi-Fi 6 with speeds of 2402 Mbps (5 GHz) and 574 Mbps (2.4 GHz), 4 Gigabit LAN ports and TP-Link HomeShield security features to make it more reliable and safe. Next, Archer AX73 V2 is priced at RM495 where it supports Wi-Fi 7 with fast speeds of 2882 Mbps (5 GHz) and 688 Mbps (2.4 GHz). Also, it can provide 1x 2.5 Gbps WAN, 1x 2.5 Gbps LAN, and 3x 1 Gbps LAN ports for good connectivity. Lastly, the Archer BE230 is priced at RM 469 and offers 4804 Mbps (5 GHz) and 574 Mbps (2.4 GHz) Wi-Fi 6. Featuring 2 x 2.5 Gbps ports and 3 x 1 Gbps LAN ports, ensuring excellent connectivity and security features such as SPI firewall, access control, and IP and MAC binding. After considering the Faculty's needs, we decided that Archer BE230 was selected for its high performance, scalability, and cost-effectiveness.

Product	TP-Link Archer AX55 WL AX3000	Archer BE230 BE3600 Dual-Band	TP Link Archer AX73 AX5400 Dual Band
Specification	• Wi-Fi Standard: Wi-Fi 6 • Wi-Fi Speed: 5GHz: 2402 Mbps, 2.4 GHz: 574 Mbps	• Wi-Fi Standard: Wi-Fi 7 • Wi-Fi Speed: 5GHz: 2882 Mbps, 2.4 GHz: 688 Mbps	• Wi-Fi Standard: Wi-Fi 6 • Wi-Fi Speed: 4804 Mbps (5 GHz), 574 Mbps (2.4 GHz)

	• Ethernet ports: 4 x Gigabit LAN, 1x Gigabit WAN • Security Features: TP-Link HomeShield	• Ethernet ports: 2x 2.5G Ports: 1x 2.5 Gbps WAN port, 1x 2.5 Gbps LAN port and 3x 1 Gbps LAN ports • Security Features: SPI Firewall, Access Control, IP & MAC Binding, and etc.	• Ethernet ports: 2x 2.5G Ports: 1x 2.5 Gbps WAN, 1x 2.5 Gbps LAN and 3x 1 Gbps LAN • Security Features: SPI Firewall, Access Control, IP & MAC Binding, and etc.
Price	RM 299	RM 469	RM 495
Reference	https://www.harveynorman.com.my/computing/networking/routers-and-extenders/tp-link-archer-ax55-ax3000-dual-band-gigabit-wi-fi-6-router.html	https://czone.my/czone/tp-link-archer-be230-wifi-7-router-dual-band-be3600.html?srsId=AfmBOorBZdSXzGtOmovUNroI_G8tNyV6PaqkoQ-sNRdQXIL8ELJKPuOP	https://www.berdaya2u.com.my/tp-link-archer-ax73-ax5400-dual-band-gigabit-wi-fi-6-router

3. Server

A server is a computer or system that provides resources, data, services, or programs to other computers, known as clients, over a network. In theory, whenever computers share resources with client machines they are considered servers. There are many types of servers, including web servers, mail servers, and virtual servers. Thus, for our labs setup, we chose the most suitable four servers which are Cisco UCS C240 M5, Cisco UCS C240 M3, Dell Power Edge R620 and Cisco Blade Server UCS B420 M3. Then, make a comparison between them and decide that the Cisco UCS C240 M5 as our lab server. This is due to the fact that its robust specs make it a future-proof solution capable of handling both current and anticipated needs of a lab environment. Firstly, it supports for up to two high-core-count Intel Xeon processors, the M5 can handle heavy multitasking, making it ideal for labs running virtual machines, simulations, or data-intensive tasks. Besides that, its ability to scale up to 3 TB of DDR4 RAM means the M5 can handle a large number of concurrent users or virtual machines. This is advantageous for labs that need to run multiple instances or support virtualization. With 24 hot-swappable bays, it provides high flexibility in configuring storage according to lab needs. Dual 10GBASE-T Ethernet ports offer faster network speeds, which is ideal for labs with high data transfer requirements. Lastly, the Cisco UCS C240 M5 offers excellent value, providing advanced features, scalability, and power at a lower price than some less capable options in the list.

Product	Cisco UCS C240 M5	Dell Power Edge R620	Cisco Blade Server UCS B420 M3
Specification	• Processor supports up to 2 scalable Intel Xeon processors • RAM is 3TB	• Processor is Intel Xeon E5-2603 • 4 cores • RAM is 8GB DDR3	• Processor type is Up to four Intel Xeon E5 v2 • Up to 3TB depending on

	DDR4 - Storage is 24 hot-swappable HDD/SSD bays (supports SAS or SATA drives), though no drives or trays are included by default - Power supply is dual hot-swappable PSUs - Networking: dual 10GBASE-T Ethernet ports, plus additional network interfaces	- No HDD - Power supply is 495W PSU	memory type, 48 DIMM slots, - RAM is 512GB DDR3 - Storage is 512GB - RAID controller is LSI 2208R
Price	RM 1329.91	RM 2200.00	RM 2898.71

Reference	https://www.serverschmiede.com/konfiguration_bulk/en/cisco-c240-m5-24x-25-sff-2x-intel-xeon-scalable-lga3647-ddr4-ecc-raid-2x-psu-2u-19-rack-server-cto-?srsltid=AfmBOorKQXjT2Uz-pWznSNrwRGhYP2HLSsECS8B06QepY4pzlCwgukVQrTQ	https://fishbone.com.my/server/refurbished-dell-power-edge-r620-server-no-hdd?srsltid=AfmBOoqg56NB1b1Y5KU7GC PbJTZZB5ahPPji7C CfigUwDKzRm3XBBX8p8W8	https://compan-it.com/en/server/1173477752-cisco-blade-server-uc-s-b420-m3-4sff-2x-xeon-e54620-v2-512-gb-ram.html?srsltid=AfmBOooDjuW_sB4VcbmihJxQCQLzzsl_QgVgM3TdMIT7ALWYyCpl9BrJCEI
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4. Cable

A cable, or network cable establishes the connection that enables the transfer of data between devices in a network. For high-speed Local Area Network (LAN) connections, Category 6 (Cat6) cables are commonly used due to their ability to support Gigabit Ethernet and high bandwidths. When selecting network cables, we will consider and compare the types of cable, and its materials, length, and cost. For our network setup, we have found three Cat6 cables that suit our requirements: DINTEK CAT6 Cable, Belden USA Original Cat6 UTP Network Cable, and PROLiNK Cat6 23AWG Solid Bare Copper UTP Network Ethernet Cable. Out of the three the DINTEK CAT6 Cable is the most affordable, priced at RM508 per 305m. The Belden USA Original Cat6 Cable is slightly pricier at RM587, while the PROLiNK Cat6 Cable is the most expensive, priced at RM699 per 305m. All of the cables use Unshielded Twisted Pair (UTP) technology and are made of copper to ensure stable performance. The DINTEK CAT6 Cable is made out of pure annealed copper, while the Belden USA Original Cat6 UTP Network Cable and PROLiNK Cat6 23AWG Solid Bare Copper UTP Network Ethernet Cable uses solid bare copper. From the comparison, we decided to choose the Belden USA Original Cat6 UTP Network Cable as it is reliable with its solid bare copper material and is also affordable in terms of cost.

Product	DINTEK CAT6 Cable / UTP Network Cable 24AWG	Belden USA Original Cat6 UTP Network Cable	PROLiNK Cat6 23AWG Solid Bare Copper UTP Network Ethernet Cable
Specification	• Type: Unshielded Twisted Pair (UTP) • Length: 305m • Material:	• Type: Unshielded Twisted Pair (UTP) • Length: 305m • Material:	• Type: Unshielded Twisted Pair (UTP) • Length: 305m • Material:

	Pure annealed copper	Bare copper	Bare copper
Price	RM508	RM587	RM699
Reference	https://martello.my/product/cat6-cable/	<a href="https://m.avio.com.my/index.php?ws=s
howproducts&prod
ucts_id=3980031&
cat=CABLE-POW
ER-ACCESSORIE
S">https://m.avio.com.my/index.php?ws=s howproducts&prod ucts_id=3980031& cat=CABLE-POW ER-ACCESSORIE S	<a href="https://shopee.com.my/Hotdeal-PROLi
NK%C2%AE-Cat6
-23AWG-Solid-Bar
e-Copper-UTP-Net
work-Ethernet-Cabl
e-(305m)-Ethernet-
Cable-LAN-Cable-
Utp-Cable-i.13127.
2913270120?sp_atk
=379bb3c1-434d-4c
5d-8dc6-a50a3b317
8ca&xptdk=379bb3
c1-434d-4c5d-8dc6
-a50a3b3178ca">https://shopee.com.my/Hotdeal-PROLi NK%C2%AE-Cat6 -23AWG-Solid-Bar e-Copper-UTP-Net work-Ethernet-Cabl e-(305m)-Ethernet- Cable-LAN-Cable- Utp-Cable-i.13127. 2913270120?sp_atk =379bb3c1-434d-4c 5d-8dc6-a50a3b317 8ca&xptdk=379bb3 c1-434d-4c5d-8dc6 -a50a3b3178ca

5. Wireless access point

A wireless access point (WAP) is a network device that allows wireless-enabled devices to connect to a wired network. Therefore, for the Faculty of Computing's new building, the TP-Link EAP653 AX3000 Ceiling Mount WiFi 6 Access Point is the most suitable wireless access point. This device offers ultra-fast true Wi-Fi 6 speed of 2976 Mbps with advanced features like 1024-QAM, HE60 and Long OFDM Symbol for higher data rates. Not only that, it's integrated with Omada SDN (Software Defined Networking) to allow centralized management which aligns with the Faculty of Computing's need for a system that is easy to manage. Also, it supports Mesh Wi-Fi and the price is RM 576 which is affordable, because it provides excellent features compared to other WAP, such as Cisco Catalyst 9115AX Series Access Point (C9115AXI-K) that just offering MU-MIMO and OFDMA, is more expensive at RM 3309.99. In addition, even though the TP-Link AC1350 is cheaper, it limits the speed by 1317 Mbps which is lower speeds, making it less ideal for handling high-performance requirements. Thus, the TP-Link EAP653 AX3000 balances cost-effectiveness with cutting-edge technology to meet the needs of labs, hybrid classrooms, video conferencing rooms, and student lounges.

Product	TP-Link AC1350 Wireless MU-MIMO Gigabit Ceiling Mount Access Point	TP-Link EAP653 AX3000 Ceiling Mount WiFi 6 Access Point	Cisco Catalyst 9115AX Series Access Point (C9115AXI-K)
Specification	• Max speed: 1317 Mbps • Signal rate: 450Mbps on 2.4GHz and 867Mbps on 5GHz • MU-MIMO for simultaneously	• Max speed: 2976 Mbps • Ultra-Fast True Wi-Fi 6 Speeds • 1024-QAM, HE60 and Long OFDM Symbol for higher data rates.	• High-performance, specific max speed not mentioned but supports Wi-Fi 6 standards • Include MU-MIMO (Multi-User, Multiple-Input, Multiple-Output)

	transfer data with multiple devices Secure guest network along with multiple authentication options.	- Integrated into Omada SDN (Software Defined Networking) - Supports Mesh WiFi	and OFDMA (Orthogonal Frequency-Division Multiple Access) Advanced security to protect against network breaches.
Price	RM 659	RM 576	RM 3309.99
Reference	https://www.sysnapse.com/tp-link-ac1350-wireless-mu-mimo-gigabit-ceiling-mount-access-point-5ghz-5dbi?gad_source=1&gclid=Cj0KCOiA0fu5BhDQARIsAMXUBOKxmGbhyWPJd8izyIMQeyzeFpPwDgnYfutOe85Otru-ESFOnKIZWpgaAgDeEALw_wcB	https://www.networkhardwares.com/en-my/products/tp-link-eap653-tp-link-ax3000-ceiling-mount-dual-band-wi-fi-6-access-point-eap653?variant=41654228123853&utm_source=google-ads&utm_campaign=&utm_agid=&utm_term=&creative=&device=c&placement=&gad_source=1&gclid=Cj0KCOiA0fu5BhDQARIsAMXUBOL7ToDj54WbavMwyAW9oqIFMqfLq1sylo6quHEYipa5pNg9-WraWalaAoVhEALw_wcB	https://www.sysnapse.com/Cisco-Catalyst-9115AX-Series-Access-Point-C9115AXI-K-with-Access-License?gad_source=1&gclid=Cj0KCOiA0fu5BhDQARIsAMXUBOLxiFMEu15aNAH3WbT_Ev4gsorlG8cBXC H66002ISLuljkh9b8IL5EaAuYwEALw_wcB

6. Firewall

A firewall is a network security device that separates a trusted internal network from an external network deemed untrustworthy, such as the internet. It regulates incoming and outgoing network traffic based on preset security rules. Firewalls are paramount in shielding networks from unauthorized access, harmful activities, and potential threats, and can exist as hardware, software, software-as-a-service (SaaS), or public or private (virtual) cloud. Our lab firewall should provide high throughput which is at least 1 Gbps to handle heavy traffic, support a large number of concurrent sessions like 500,000 or more, and allow for multiple VLANs for network segmentation. It should also include features like intrusion prevention, content filtering, and VPN support for secure remote access. Scalability is key, with the ability to handle new connections per second (10,000 or more) and provide robust management and reporting tools. Additionally, support for role-based access control and real-time threat detection ensures a secure and efficient environment for academic activities.

Product	FortiGate 30E	ASA5550-BUN-K9	ASA5506-K9
Specification	- Internal storage is 16 GB - Firewall throughput is 950 Mbps - Concurrent sessions (TCP) is 900,000 - Firewall latency) 64-byte UDP packets) is	- Memory is 4 GB - Firewall throughput is up to 1.2 Gbps - Concurrency connection is 650,000 - Virtual Interfaces (VLANs) is 250 - New	- Memory is 4 GB - Stateful inspection throughput (multiprotocol) is 300 Mbps - Concurrent sessions are 20,000 - 50,000 - Virtual interfaces (VLANs) are

	130 μs • New sessions/second (TCP) is 15,000 • IPsec VPN throughput (512 byte packets) is 75 Mbps	connections are 650,000 • IPsec VPN peers are 5000	5, 30 with security plus license • New connections per second is 5,000 • IPsec site-to-site VPN peers is 10, 50 with security plus license
Price	RM 397.96	RM 2079.25	RM 3554.84
Reference	https://www.hi-network.com/used-fortigate-30e-fortinet-firewall-s-1ge-ports-use-for-study-vpn.html?srsltid=AfmBOorYy1WeW-LBDFjTqfdiWbJ8YA-DGCMNN89GxjQb7DGMJZ6wjmi-i9Q	https://www.hi-network.com/used-asa5550-bun-k9-cisco-8ge-ports-unlimited-users-firerwalls.html?srsltid=AfmBOoqsQLFpaYMTKkG8GoDE-CTt8rq5cq_OPYISylK4io05Ae2O9NhPu8k	https://www.hi-network.com/asa5506-k9-unlimited-users-cisco-firerwalls-100-200-300-etc.html?srsltid=AfmBOoqGXg0pPfeDNbFQtg1Ec_xgxRP_o5LzLODcikWd-PQM604H4ukIoOY

7. PC

PC, or personal computer is a computer meant for individual use. It is typically used for all sorts of tasks such as browsing the internet, entertainment, and communication. For students of Faculty of Computing, they will have to use PCs to perform intensive tasks such as programming and software development. Thus it is important to select a PC that has high performance while being cost efficient. Three PCs that we have found are the Acer Aspire XC-1785-14700W11S, HP Pro Tower 400 G9 PCI Desktop PC, and Dell OptiPlex 7080 SFF Business Desktop PC. All three of the PCs have an Intel Core i7 processor, which provides excellent multi-core performance compared to other PCs in the market. The Acer Aspire XC-1785-14700W11S features a 16 GB DDR5 memory which is superior to the 8 GB DDR5 and 16 GB DDR4 memory of HP Pro Tower 400 G9 PCI Desktop PC and Dell OptiPlex 7080 SFF Business Desktop PC respectively. In terms of storage, the Acer Aspire XC-1785-14700W11S also wins with a 1024GB storage compared to the others. It also runs Windows 11 Home and has Intel UHD Graphics with are sufficient for computing tasks. Thus, we decided that the Acer Aspire XC-1785-14700W11S is the most suitable option for its high performance and reasonable price.

Product	Acer Aspire XC-1785-14700W11S	HP Pro Tower 400 G9 PCI Desktop PC	Dell OptiPlex 7080 SFF Business Desktop PC
Specification	- Processor: Intel® Core™ i7 14700 - Memory: 16 GB DDR5 - Storage: 1024GB M.2 2280 PCI-e Gen4	- Processor: 14th Generation Intel® Core™ i7 processor - Memory: 8 GB DDR5-4800 MHz RAM	- Processor: Intel Core i7-10700 10th Gen Octa-Core Processor - Memory: 16 GB DDR4 2933 MHz DDR4

	SSD - Operating system: Windows 11 Home - Graphics: Intel® UHD Graphics	- Storage: 512 GB SSD Hard drive - Operating system: Windows 11 Pro - Graphics: Intel® UHD Graphics 770	- Storage: 256 GB M.2 512GB PCIe NVMe Class 40 Solid State Drive - Operating system: Windows 10 Professional 64 Bit English Edition - Graphics: Integrated Intel® HD Graphics 630
Price	RM4399	RM4465	RM6426
Reference	https://store.acer.com/en-my/acer-consumer-desktop-acer-aspire-xc-series-xc-1785-14700w11s-intel-core-i7-14700	https://www.hp.com/my-en/shop/hp-pro-tower-400-g9-pci-desktop-pc-a3mk9pt.html?facetref=f5d119318cd60c3b	https://shopee.com.my/Dell-OptiPlex-7080-SFF-Business-Desktop-PC-I7-10700-16GB-RAM-256GB-SSD-Win10-Pro-3-Years-Dell-Warranty-i.84872933.8618147894?sp_atk=

			9620612f-d6a0-4d7d-bb37-d7b67b9333a9&xptdk=9620612f-d6a0-4d7d-bb37-d7b67b9333a9
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8. Projector

A projector is an electronic device that displays images, videos, or presentations on a screen or wall. Projectors work by projecting light through a lens that magnifies the image. Projectors are commonly used by lecturers for teaching, and students for presentations in classrooms, conference rooms, labs, and other environments. Projectors can display content from various sources such as computers, tablets or media players, and often have features such as different resolutions, brightness levels, and connectivity options to suit different viewing environments. We compared three projectors which are Casio XJ-A142, BenQ MX560, and Epson EB-X51. First, the RM2,699 Casio XJ-A142 offers 2,500 lumens of brightness with a lamp life of up to 20,000 hours. It supports XGA resolution and has a relatively low contrast ratio of 1800:1 which makes it suitable for environments that require high brightness and long lamp life. In addition, the BenQ MX560, priced at RM1,699, is even brighter at 4,000 lumens with a lamp life of 6,000 to 15,000 hours. With a contrast ratio of up to 20,000:1, it's a strong contender for clear, bright images in bright rooms. Finally, the Epson EB-X51 is priced at RM2,059 and has a brightness of 3,800 lumens and a contrast ratio of 16,000:1. In energy-saving mode, the lamp life is extended to 12,000 hours. After considering the building's requirements for brightness, energy efficiency and adaptability, the Epson EB-X51 was chosen for its optimal balance of performance, functionality and value.

Product	Casio XJ-A142 LED 2500 Lumens Projector	BenQ MX560 4000 Lumens XGA DLP	Epson EB-X51 XGA 3800 Lumens 3LCD Projector
Specification	- Brightness: 2500 Lumens - Lamp Life: up to 20000 hours - Contrast ratio: 1800:1 - Resolution:	- Brightness: 4000 Lumens - Lamp Life: Normal 6000 hrs, ECO 10000 hrs, SmartEco 10000 hrs,	- Brightness: 3,800 lumens - Lamp Life: up to 12,000 hours in ECO Mode - Contrast ratio: 16,000: 1

	Support XGA (1024 x 768) Power consumption: 150W if Eco mode “off” is selected, and 115W for Eco mode “Eco 1”	LampSave 15000 hrs - Contrast ratio: 20,000:1 - Resolution: Support VGA(640 x 480) to WUXGA_RB(1920 x 1200) - Power consumption: 260W	- Resolution: Support native XGA - Power consumption: 327W/225W if lamp in normal/eco mode, for standby (network/ energy saving) is 2.0W/0.3W
Price	RM 2699	RM 1699	RM 2059
Reference	https://www.projector.com.my/casio/casio-xj-a142-led-2500-lumens-projector	https://projectordisplay.com/product/benq-mx560-4000-lumens-xga-dlp-business-projector/?gad_source=1&gclid=Cj0KCQiA0fu5BhDQARIsAMXUBOLKRplPPGxcUYDdxH4cS_8KI7oumH9e_pT6sqDnXibLakIWdxcEUZcaAqsNEALw_wcB	https://projector.my/epson-eb-x51-xga-3800-lumens-3led-projector

9. Printer

A printer is a device that accepts text and graphic output from a computer and transfers the information to paper, usually to standard-size, 8.5" by 11" sheets of paper. Printers vary in size, speed, sophistication and cost. In general, more expensive printers are used for more frequent printing or high-resolution color printing. There are many different printer manufacturers available today, including Canon, Epson, Hewlett-Packard, Xerox and Lexmark. There are also several types of printers such as inkjet printers, laser printers, 3D printers, photo printers and so on. Thus, for our faculty, the main specifications for printers that we will consider are color, resolution, and speed. As the comparison table is shown below, we will select HP LaserJet Enterprise MFP M430f as our choice because it is more suitable and fulfills our requirements.

Product	HP OfficeJet Pro 9730	Brother DCP-L2640DW	HP LaserJet Enterprise MFP M430f
Specification	● Print speed (black ISO) up to 22 ppm ● Print speed (black, draft, letter/A4) up to 34 ppm ● Print speed (color, ISO) up to 18 ppm ● Print quality (black, best) up to 1200x1200 rendered dpi ● Print quality	● Print speed up to 34/36 ppm ● Print resolution up to 1200x1200 dpi ● Printer type is a mono laser printer ● Memory capacity is 256 MB ● Connectivity is Hi-Speed USB 2.0,	● Print speed (black) up to 40 ppm ● Print resolution up to 1200x1200 dpi ● A ● Memory is 2GB ● Wireless capability is Optional, enabled with purchase of a

	(color, best) up to 4800x1200 optimized dpi on HP advance photo paper • Print technology is HP thermal inkjet • File formats supported are pdf, tiff, jpeg • Memory is 512 MB • Wireless capability with built-in dual-band Wi-Fi 802.11 a/b/g/n/ac and self-heal solution	Ethernet 10Base-T/100 Base-TX, IEEE 802.11b/g/n (Infrastructure Mode), IEEE 802.11g/n (Wi-Fi Direct)	wireless hardware accessory. • Connectivity is 2 Hi-Speed USB 2.0 Host; 1 Hi-Speed USB 2.0 Device; 1 Gigabit Ethernet 10/100/1000T network; 1 Fax
Price	RM 1022.84	RM 919.01	RM 4013.00
Reference	https://mtech-services.com/product/hp-officejet-pro-9730-wide-format-all-in-one-printer-print-copy/?srsltid=AfmBOoqH4rVc1RznjsU4-fSy	https://global.microless.com/product/brother-dcp-l2640dw-com-pact-mono-laser-multi-function-printer-print-scan-copy-up-to-34-	https://www.hp.com/my-en/shop/hp-laserjet-enterprise-mfp-m430f-3pz55a.html#specs

	dbxh-a7iUOtmQI6RLq lulityCuLCK0UeBxI	36-ppm-a4-ltr-print-s peed-1200-x-1200-dp i-resolution-2-lines-lc d-display-black-dcp-l 2640dw/?currency=u sd&srsltid=AfmBOo o-vHMXWplentgA5 dTGWB7HR94oPtk3 TijnQCAFKPGOEDy -LZkOx5U	
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3.3.2 Expected cost

No	Device	Quantity	Price/each (RM)	Total Price (RM)
1.	Ubiquiti USW-Pro-48-PoE	9	5779	52011.00
2.	Archer BE230 BE3600 Dual-Band	10	469	4690.00
3.	Cisco UCS C240 M5	10	1329.91	13299.10
4.	Belden USA Original Cat6 UTP Network Cable	8	587	4696.00
5.	TP-Link EAP653 AX3000 Ceiling Mount WiFi 6 Access Point	4	576	2304.00
6.	ASA5550-BUN-K9	8	2079.25	16634.00
7.	Acer Aspire XC-1785-14700W11S	160	4399	703840.00
8.	Epson EB-X51 XGA 3800 Lumens 3LCD Projector	6	2059.00	12354.00
9.	HP LaserJet Enterprise MFP M430f	4	4013.00	16052.00
Total cost				826153.10

3.3.3 Reflection

1. Are you surprised by the prices? How were you surprised?

Yes, we are surprised by the prices of the devices required. In the process of surveying the market for suitable devices, we discovered that most of them have a large price range, with the most expensive ones reaching up to tens of thousands. Since we are constructing a large scale building with extensive network architecture, we need to acquire a large quantity of the same devices at once. For example, a floor needs an estimate of 80 personal computers. When added up it produces a large amount of money which we are surprised with. Thus, this shows that it is important for us to find devices for our building that are balanced in terms of performance, quality, and cost. Besides, we also have to closely monitor our budget and make sure that the total costs for all devices are within it.

2. Have you ever considered cost as a factor for choosing networking devices?

Yes, cost is an important factor when choosing networking equipment. This is because we have a limited budget and we need to make sure we stay within the available budget. Therefore, the way to choose the right network equipment is to determine the required functions. In this way, we have a direction of choice, instead of blindly thinking that the most expensive is the best. However, when cheap equipment is available, we must recognize the brand. If it is not a well-known brand, do not buy it because problems such as poor performance and short lifespan may occur. Therefore, comparing devices carefully is one of the most effective solutions. This way the network can run well and the budget can be well controlled.

3. What are the major differences between the same devices from different brands? For example, Cisco and Huawei Routers.

Reputation, design philosophy, performance, features, cost, and compatibility are often the main areas where devices from various brands differ from one another. The trust and dependability that a brand has developed over time is reflected in its reputation. Some companies are recognized for their inventiveness, while others prioritize price. Usability and scalability are influenced by design philosophy. Some of the manufacturers prioritize high-end enterprise functionality and modularity, while others prioritize

cost-effectiveness and simplicity. Speed, dependability, and scalability are all impacted by performance, which varies according to software optimization and hardware quality. While some manufacturers take the lead in implementing new standards, others only provide basic capabilities. This results in differences in features and protocol support. Pricing schemes that include license fees and upfront expenses can have a big effect on the overall cost of ownership. Last but not least, compatibility with different ecosystems and devices varies, with certain brands excelling at seamless integration.

3.4 Task 4 - Making The Connections - LAN and WAN

3.4.1 Work areas on the floor plan

3.4.1.1 Ground Floor

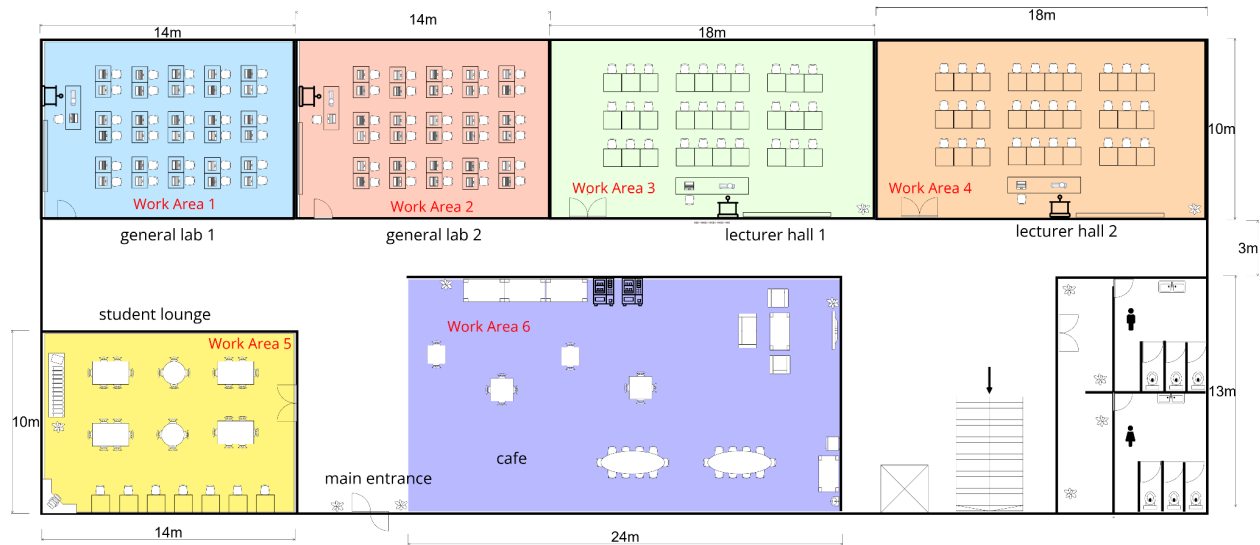


Figure 3: Work Area on the Ground Floor

On the ground floor, there are 6 work areas, which consists of General Lab 1, General Lab 2, Lecturer Hall 1, Lecturer Hall 2, the student lounge, and the cafe. There are 30 workstations in both General Lab 1 and General Lab 2. Each workstation is connected to switches that link to a centralized router, which provides a smooth network connection for students to carry out any practical activities. Besides, Lecturer Hall 1 and Lecturer Hall 2 are equipped with audio and visual equipment to support lectures effectively. There is also a student lounge that has a fast internet connection for students to rest and complete their work comfortably outside of lecture times. Lastly, the cafe with tables and chairs provide a place for students to have refreshments while carrying out social interactions such as discussions.

3.4.1.2 First Floor

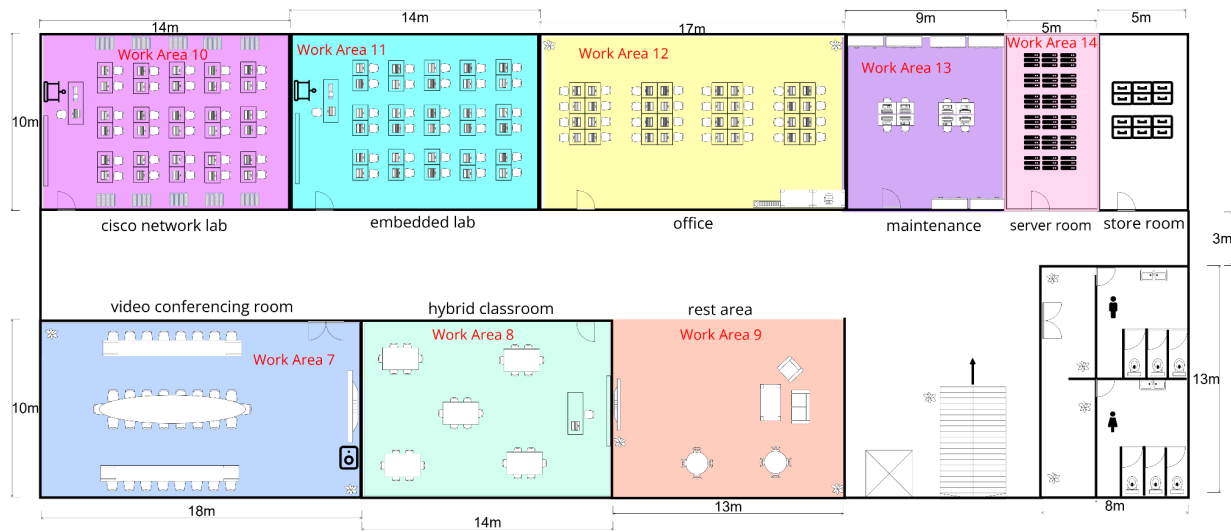


Figure 4: Work Area on First Floor

For the first floor, there are 8 work areas, which is the Cisco network lab, the embedded lab, office, maintenance room, server room, video conferencing room, hybrid classroom, and a rest area. All of the labs on this floor have 30 workstations that are connected to a local network switch, which is linked to the building's routers and servers. The office is used by administrative staff to manage student related matters. Next, there is also a maintenance room that houses diagnostic tools and spare parts for IT equipment. Admins and technicians can use this space to manage and fix the building's network infrastructure. Lastly, the server room contains servers, routers, switches, and backup systems. All workstations on the ground floor and first floor are connected to the servers via a hierarchical network design.

3.4.2 Network Diagram

3.4.2.1 Ground Floor

3.4.2.1.1 Lecturer Hall

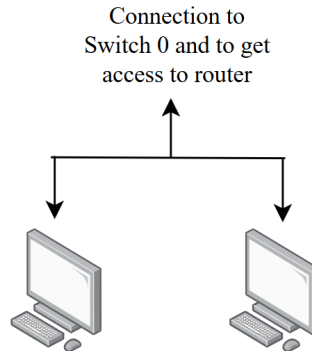


Figure 5: Lecturer Hall Network Diagram

3.4.2.1.2 General Lab

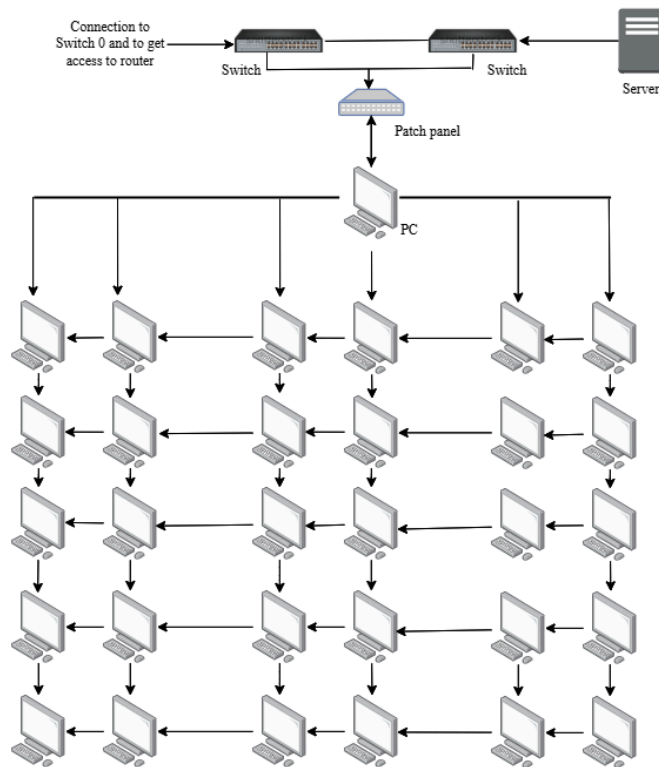


Figure 6: General Lab Network Diagram

3.4.2.1.3 Student Lounge

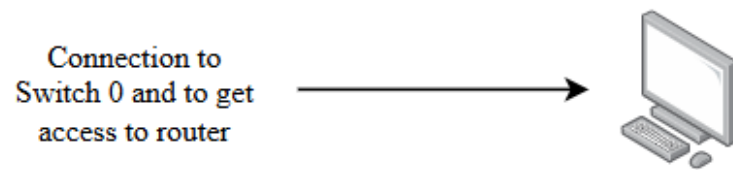


Figure 7: Student Lounge Network Diagram

3.4.2.1.4 Cafe

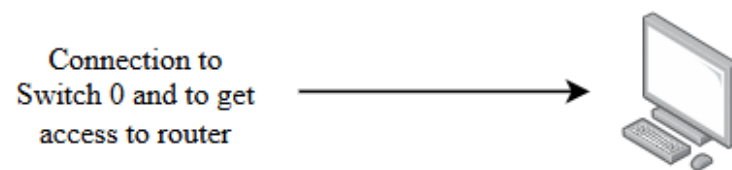


Figure 8: Cafe Network Diagram

3.4.1.2 First Floor

3.4.1.2.1 Video Conferencing Room

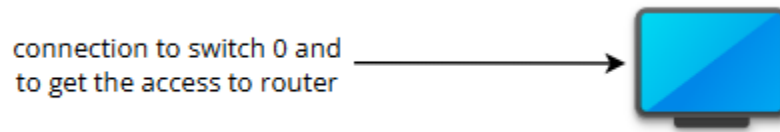


Figure 9: Video Conferencing Room Network Diagram

3.4.1.2.2 Hybrid Classroom

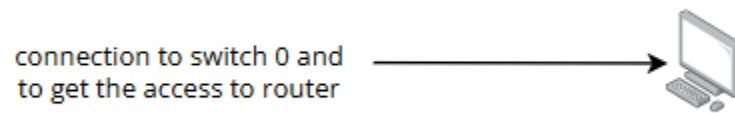


Figure 10: Hybrid Classroom Network Diagram

3.4.1.2.3 Rest Area

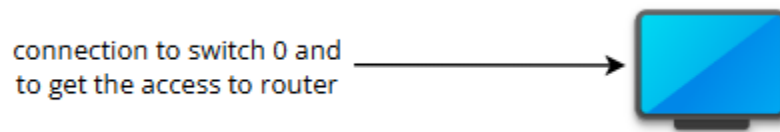


Figure 11: Rest Area Network Diagram

3.4.1.2.4 Server Room



Figure 12: Server Room Network Diagram

3.4.1.2.5 Office

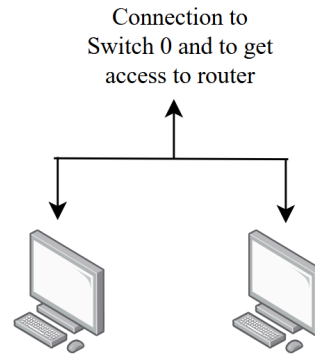


Figure 13: Office Network Diagram

3.4.1.2.6 Cisco Network Lab

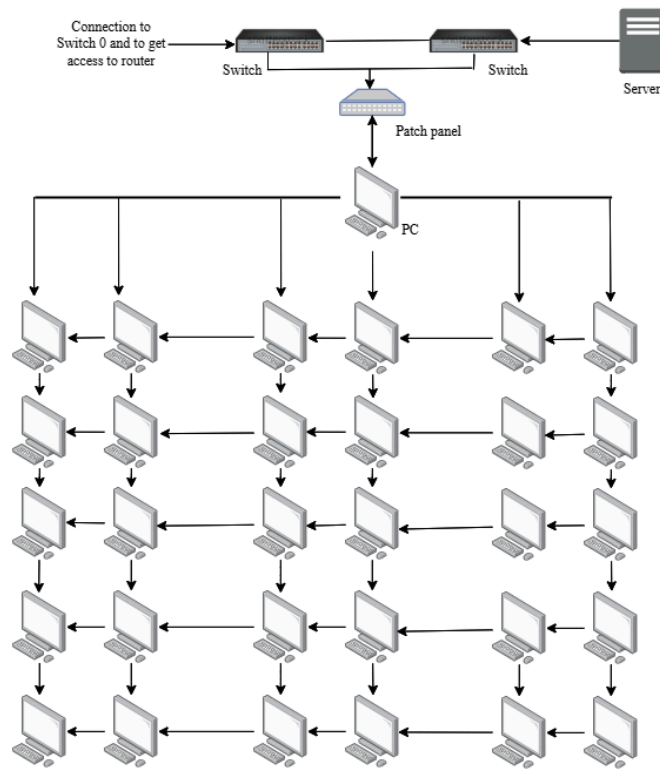


Figure 14: Cisco Network Lab Network Diagram

3.4.1.2.7 Embedded Lab

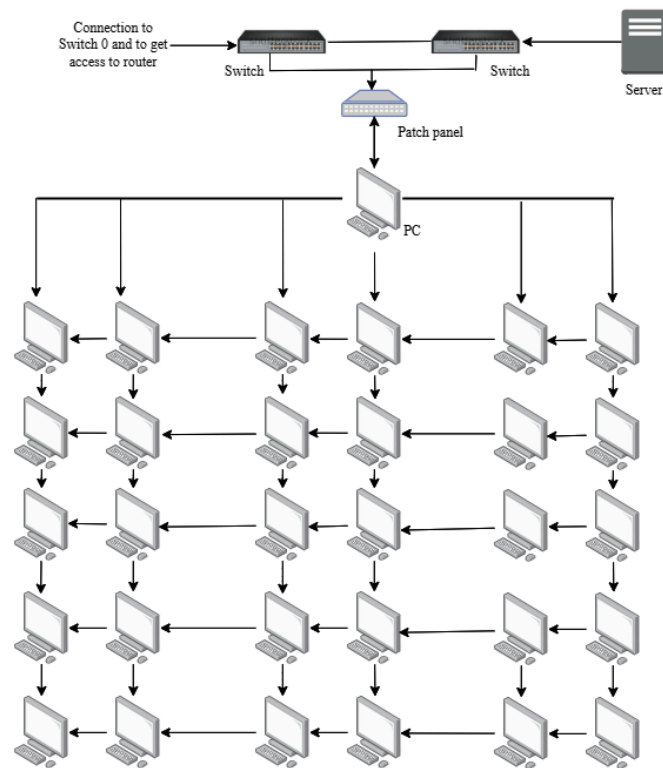


Figure 15: Embedded Lab Network Diagram

3.4.1.2.8 Maintenance Room



Figure 16: Maintenance Room Network Diagram

3.4.3 Cables & Connections

3.4.3.1 Floor Plan

3.4.3.1.1 Ground Floor

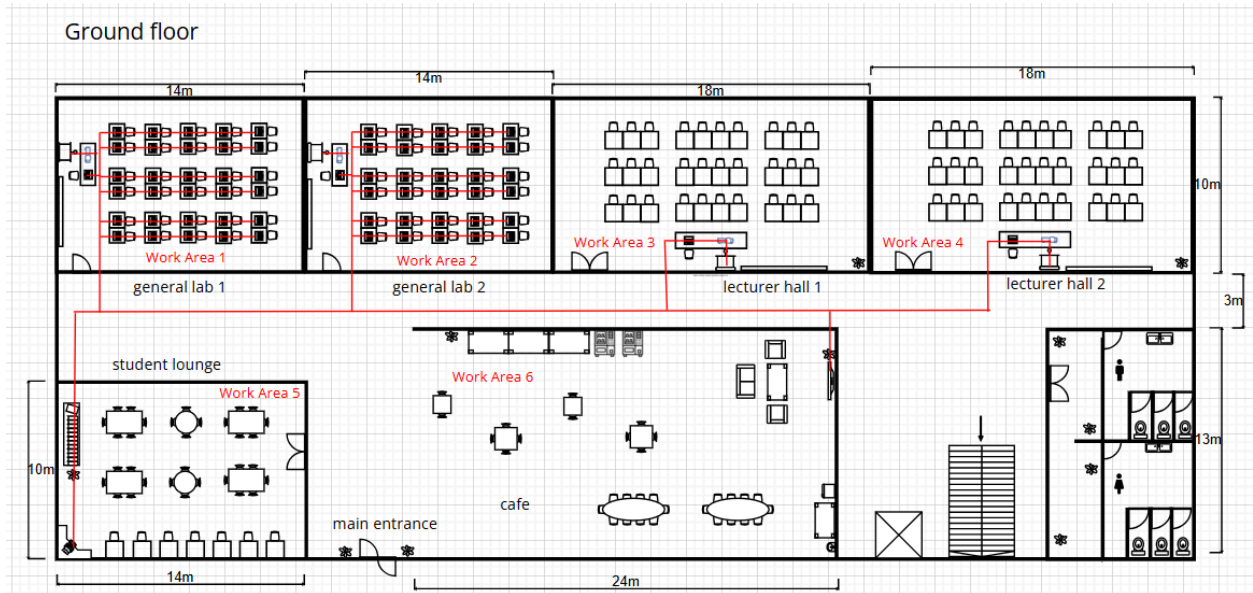


Figure 17: Cable distribution of ground floor

3.4.3.1.2 First Floor

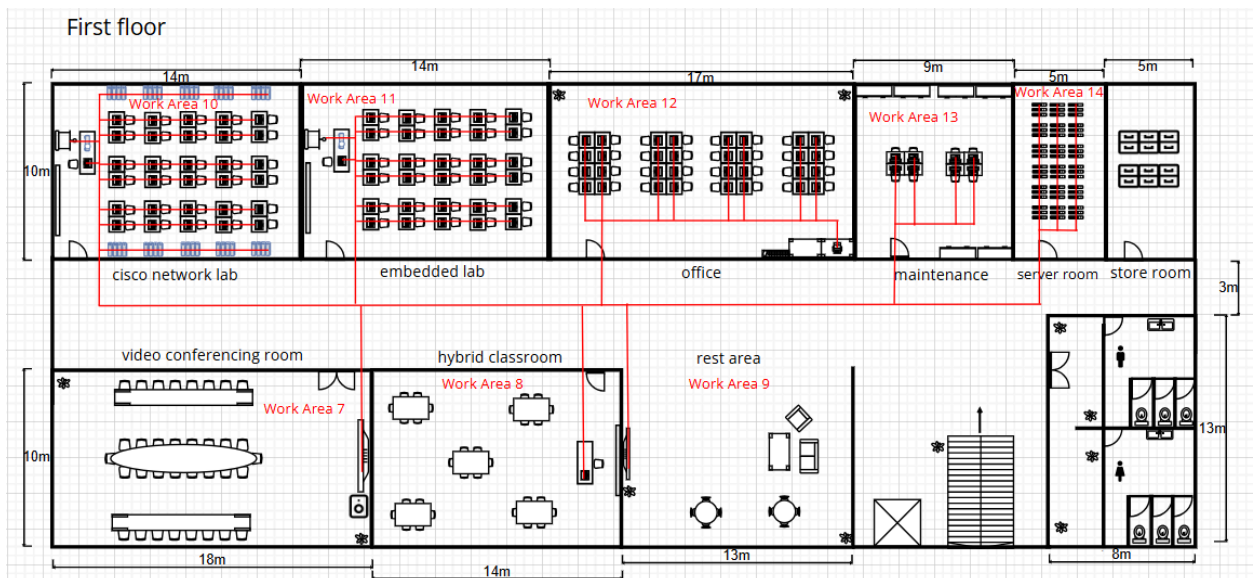


Figure 18: Cable distribution of the first floor

3.4.3.2 Number of connections, patch cords and switch ports needed

Area	Number of Switch Ports Needed
General Lab 1	32
General Lab 2	32
Lecturer Hall 1	2
Lecturer Hall 2	2
Student Lounge	1
Cafe	1
Video Conferencing Room	1
Hybrid Classroom	1
Rest Area	1
Cisco Network Lab	42
Embedded Lab	32
Office	33
Maintenance	8
Server Room	18
Total	206

Table 3.4.3.2.1: Number of switch ports needed

Area	Number of Patch Cords Needed
General Lab 1	32
General Lab 2	32
Lecturer Hall 1	2
Lecturer Hall 2	2
Student Lounge	1
Cafe	1
Video Conferencing Room	1

Hybrid Classroom	1
Rest Area	1
Cisco Network Lab	42
Embedded Lab	32
Office	33
Maintenance	8
Server Room	18
Total	206

Table 3.4.3.2.2: Number of patch cords needed

Area	Number of RJ45 Connector Needed
General Lab 1	64
General Lab 2	64
Lecturer Hall 1	4
Lecturer Hall 2	4
Student Lounge	2
Cafe	2
Video Conferencing Room	2
Hybrid Classroom	2
Rest Area	2
Cisco Network Lab	84
Embedded Lab	64
Office	66
Maintenance	16
Server Room	36
Total	412

Table 3.4.3.2.3: Number of RJ45 connectors needed

3.4.3.3 Identify the cable length & type

Description	Cable Type	Length (m)
Ground Floor		
General lab 1	CAT 6 cable	71
General lab 2	CAT 6 cable	71
Lecturer hall 1	CAT 6 cable	27
Lecturer hall 2	CAT 6 cable	16
Cafe	CAT 6 cable	28.5
Student lounge	CAT 6 cable	26.5
Total length of the Ground Floor (m)		240
First floor		
Cisco network lab	CAT 6 cable	92
Embedded lab	CAT 6 cable	67.5
Office	CAT 6 cable	77
Maintenance	CAT 6 cable	33.5
Server room	CAT 6 cable	29
Rest area	CAT 6 cable	22
Hybrid classroom	CAT 6 cable	82
Video conferencing room	CAT 6 cable	24.5
Total length of the First Floor (m)		427.5
Fibre optic cable	Fibre optic cable	24.5

Table 3.4.3.3.1: Cable length

Based on our planning, we decided to use CAT6 cables on the ground floor and first floor of the Faculty of Computing building. These cables are used in every lab, lecturer hall, office, etc. and are used to connect servers. Therefore, to ensure a reliable and high-speed connection is produced, we chose CAT6 cables for most internal connections, and fiber optic cables for areas requiring fast data transmission. As a result, this combination delivers efficient and reliable performance for all networking needs within the building.

3.4.3.3.1 CAT 6 Cable

CAT 6 cable is chosen to be the main cable in the building of the Faculty of Computing, which is able to ensure high-speed and reliable data transmission between locations. These cables are installed in all labs, lecture halls, offices, server rooms, and others. The total length of cable routing is 240 metres for the ground floor and 427.5 metres for the ground floor. CAT 6 cables can transfer large amounts of data at high speeds of up to 500 MHz for speed-critical tasks such as file sharing and accessing learning resources.

Moreover, CAT 6 cables also reduce crosstalk because of their tighter stranded design and increased insulation. As a result, this helps ensure a more stable connection with fewer data errors. Next, the network infrastructure needs to keep up as the demand for faster internet speeds and more data usage continues to grow. Therefore, using CAT 6 cable will help cope with future technological advancements. In short, by using CAT 6 cable we can maintain a high level of connectivity and efficiency in all critical areas of the building.

3.4.3.3.2 Fiber Optic Cable

Fiber optic cables are chosen for video conference rooms that require fast and continuous data transmission. These fibre optic cables transmit optical signals, which allow data transfer speeds of up to 100 Gbps and signal transmission over the longest distances without degradation due to distance. It is therefore ideally suited to video conference rooms, where both video and voice must be transmitted clearly without any loss of quality.

The fiber optic cable runs along the perimeter of the building, connecting all video conference rooms throughout the facility to the core network. In this way, video and audio conferencing can take place without latency, making it easier for students or staff to participate in virtual meetings, online classes, or give uninterrupted presentations. In addition, fiber optic cables are future-proofed as they will not be under pressure as faculty and data demands continue to grow.

3.4.3.3.3 Wireless

Wireless networks allow devices to connect to the internet or network without using cables. Wi-Fi signals from access points allow devices to stay connected even far from the router. In the Faculty of Computing, wireless networks enable students and staff to move freely while staying online. The wireless network supports online classes, project work and resource access, ensuring strong connectivity in key areas such as lecture halls and student lounges.

3.5 Task 5 - IP Addressing Scheme

3.5.1 Subnet Mask

Subnet No	Area	IP address/ Subnet Mask (Binary)	IP address/ Subnet Mask (Decimal)
0	General Lab1 & General Lab 2	1100 0000. 0000 0000. 0000 0000. 0000 0000	192.0.0.0/11
1	Lecturer Hall 1 & Lecturer Hall 2	1100 0000. 0010 0000. 0000 0000. 0000 0000	192.32.0.0/11
2	Student Lounge & Cafe	1100 0000. 0100 0000. 0000 0000. 0000 0000	192.64.0.0/11
3	Cisco Network Lab	1100 0000. 0110 0000. 0000 0000. 0000 0000	192.96.0.0/11
4	Embedded Lab	1100 0000. 1000 0000. 0000 0000. 0000 0000	192.128.0.0/11
5	Office, Maintenance, Server room & Rest area	1100 0000. 1010 0000. 0000 0000. 0000 0000	192.160.0.0/11
6	Hybrid classroom	1100 0000. 1100 0000. 0000 0000. 0000 0000	192.192.0.0/11
7	Video conferencing room	1100 0000. 1110 0000. 0000 0000. 0000 0000	192.224.0.0/11

3.5.2 Range of IP address, network and broadcast address for the subnet

Subnet No	Area	Network Address	Broadcast Address	Range of IP Address
0	General Lab1 & General Lab 2	192.0.0.0	192.31.255.255	192.0.0.1 – 192.31.255.254 1100 0000. 0000 0000. 0000 0000. 0000 0001 – 1100 0000. 0001 1111. 1111 1111. 1111 1110
1	Lecturer Hall 1 & Lecturer Hall 2	192.32.0.0	192.63.255.255	192.32.0.1 – 192.63.255.254 1100 0000. 0010 0000. 0000 0000. 0000 0001 – 1100 0000. 0011 1111. 1111 1111. 1111 1110
2	Student Lounge & Cafe	192.64.0.0	192.95.255.255	192.64.0.1 – 192.95.255.254 1100 0000. 0100 0000. 0000 0000. 0000 0001 – 1100 0000. 0001 1111. 1111 1111. 1111 1110
3	Cisco Network Lab	192.96.0.0	192.127.255.255	192.96.0.1 – 192.127.255.254 1100 0000. 0000 0000. 0000 0000. 0000 0001 – 1100 0000. 0101 1111. 1111 1111. 1111 1110
4	Embedded Lab	192.128.0.0	192.159.255.255	192.128.0.1 – 192.159.255.254 1100 0000. 1000 0000. 0000 0000. 0000 0001 – 1100 0000. 1001 1111. 1111 1111. 1111 1110
5	Office, Maintenance, Server room & Rest area	192.160.0.0	192.191.255.255	192.160.0.1 – 192.191.255.254 1100 0000. 1010 0000. 0000 0000. 0000 0001 – 1100 0000. 1011 1111. 1111 1111. 1111 1110
6	Hybrid classroom	192.192.0.0	192.223.255.255	192.192.0.1 – 192.223.255.254 1100 0000. 1100 0000. 0000 0000. 0000 0001 – 1100 0000. 1101 1111. 1111 1111. 1111 1110
7	Video conferencing room	192.224.0.0	192.255.255.255	192.224.0.1 – 192.255.255.254 1100 0000. 1110 0000. 0000 0000. 0000 0001 – 1100 0000. 1111 1111. 1111 1111. 1111 1110

3.5.3 Range of IP address for user types based on the area

3.5.3.1 General Lab1 & General Lab 2

Users		IP Range
General Lab 1	Student (PC)	192.0.0.1 – 192.0.0.30
	Lecturer (PC)	192.0.0.31
	Projector	192.0.0.32
General Lab 2	Student (PC)	192.0.0.33 – 192.0.0.62
	Lecturer (PC)	192.0.0.63
	Projector	192.0.0.64

3.5.3.2 Lecturer Hall 1 & Lecturer Hall 2

Users		IP Range
Lecturer Hall 1	Lecturer (PC)	192.32.0.1
	Projector	192.32.0.2
Lecturer Hall 2	Lecturer (PC)	192.32.0.3
	Projector	192.32.0.4

3.5.3.3 Student Lounge & Cafe

Users	IP Range
Student Lounge	192.64.0.1
Cafe (TV)	192.64.0.2

3.5.3.4 Cisco Network Lab

Users	IP Range
Student (PC)	192.96.0.1 – 192.96.0.30
Lecturer (PC)	192.96.0.31
Projector	192.96.0.32

3.5.3.5 Embedded Lab

Users	IP Range
Student (PC)	192.128.0.1 – 192.128.0.30
Lecturer (PC)	192.96.0.31
Projector	192.96.0.32

3.5.3.6 Office, Maintenance, Server room & Rest area

Users	IP Range
Office	192.160.0.1 – 192.160.0.32
Maintenance Room	192.160.0.33 – 192.160.0.40
Server Room	192.160.0.41 – 192.160.0.55
Rest Area (TV)	192.160.0.56

3.5.3.7 Hybrid classroom

Users	IP Range
Lecturer (PC)	192.192.0.1

3.5.3.8 Video conferencing room

Users	IP Range
Presenter (TV)	192.224.0.1

Ground Floor with label

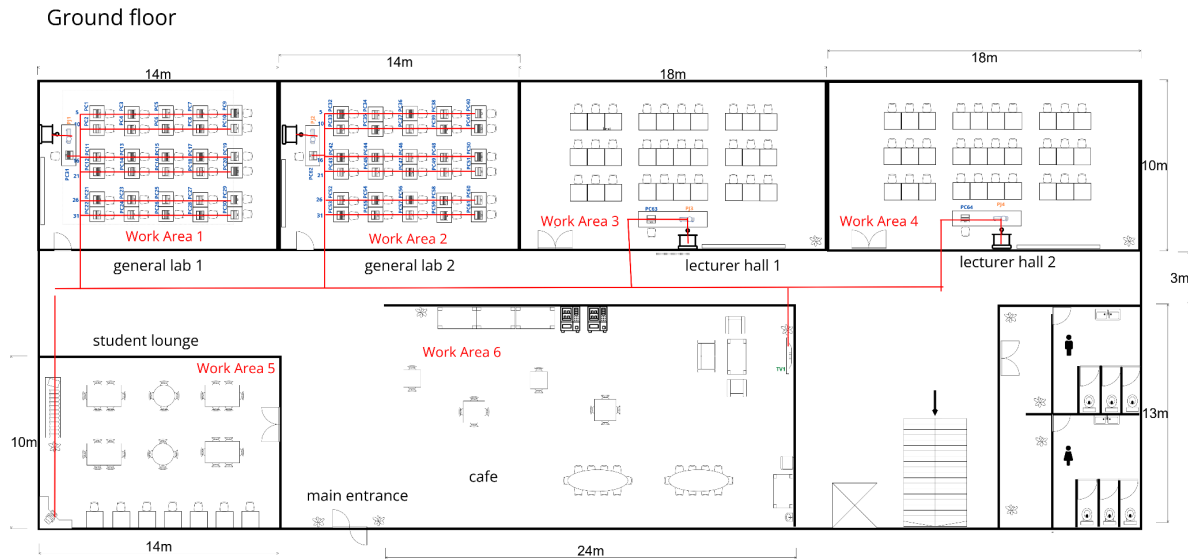


Figure 19: Ground Floor Plan with label

First Floor with label

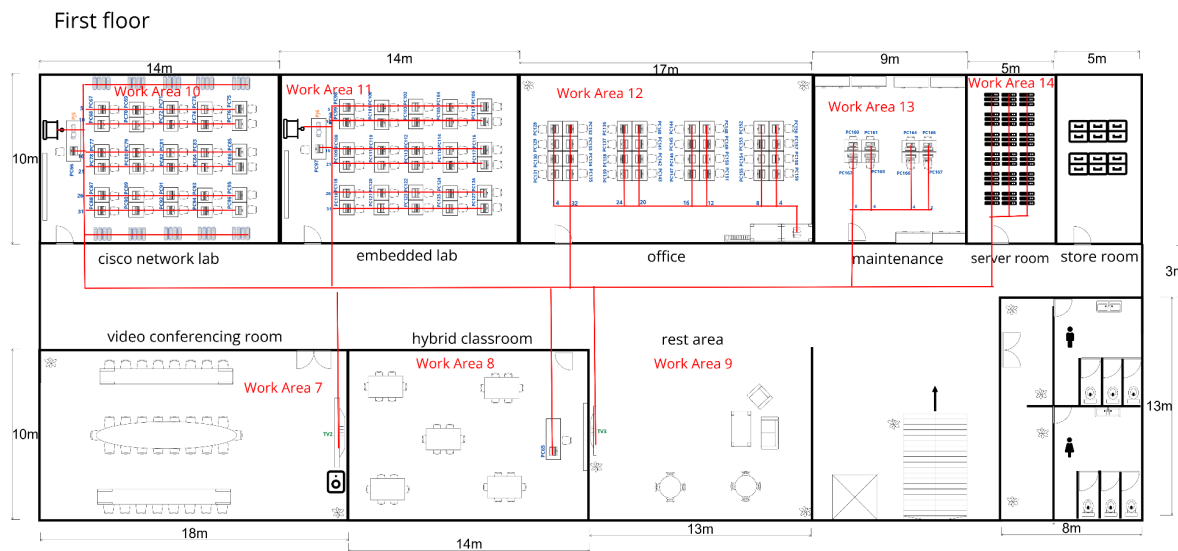


Figure 20: First Floor Plan with label

4.0 Conclusion

In conclusion, our group successfully completed all tasks from planning the network to choosing the devices and assigning IP addresses. We carefully consider the needs of our clients to ensure that network planning meets current and future needs. By working together as a team, we were able to select the right equipment to ensure that the network would perform well, be secure, and grow with future demand. Our final designs include clear layouts, wiring plans and IP addresses, all presented in a professional manner.

We work well together and divide tasks based on each person's strengths and knowledge. This allows making work more efficient. Additionally, we conducted research on network devices, IP addressing, and cabling, which helped us choose the appropriate and cost-effective devices for the network. In addition, our group was able to adjust our plans as needed and update our design with new information.

However, we faced challenges with time management, as some tasks, such as selecting devices, took longer than expected. In addition, some technical aspects, such as subnetting and IP assignment, were more complex than expected, and not all team members were familiar with these tasks. As a result, the task progressed slowly.

In order to improve our work, we can do better planning before we start the project. Having a detailed plan at the beginning of the project will help us to complete our tasks faster. Furthermore, additional network configuration and troubleshooting practice will enhance our technical skills and improve the overall effectiveness of the team.

All in all, our team did an excellent job designing the network for the new building. We learned valuable lessons in teamwork, time management, and network design, and we hope to improve these areas in future projects.

5.0 Team Members and Responsibilities

1. ALICE LEE HUI MEE

- Task 1: Background, meeting minutes
- Task 2: 3 questions and 1 feasibility
- Task 3: Router, WAP, projector, reflection question 2
- Task 4: 4 network diagram, cable length and type, meeting minutes
- Task 5: Subnet mask, range of IP address, network and broadcast address for the subnet
- Task 6: Background, conclusion, reference, compiled solution of task 5

2. CHENG KANG HUEY

- Task 1: Ground floor floor plan
- Task 2: 3 questions and 1 feasibility, meeting minutes
- Task 3: Switch, cable, PC, reflection question 1
- Task 4: Work areas on floor plan, 4 network diagram
- Task 5: Range of IP address for user types based on the area, ground floor plan
- Task 6: Abstract, team members and responsibilities, appendices, compiled solution of task 4

3. LOH HUI YI

- Task 1: First floor floor plan
- Task 2: 4 questions and 1 feasibility
- Task 3: Server, firewall, printer, reflection question 3, meeting minutes
- Task 4: 4 network diagram, cables and connections
- Task 5: First floor plan, meeting minutes
- Task 6: Introduction, compiled solution of task 1, 2, 3, table of figures

6.0 References

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Appendices

1. First Meeting Minutes

DATE/TIME	13 Oct 2024 10am		
LOCATION	Google Meet		
AGENDA	Group Name Brainstorm and Floor Layout Discussion		
Meeting MC	Alice Lee Hui Mee, Cheng Kang Huey, Loh Hui Yi		
ATTENDANCE			
NAME	TIME		REASON FOR ABSENCE
Alice Lee Hui Mee	1000 - 1130		-
Cheng Kang Huey	1000 - 1130		-
Loh Hui Yi	1000 - 1130		-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/ SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Introduction and case study review	- Alice Lee presents the case study tasks and project rubric. - All members review the file, highlighting key points and important requirements. - Discuss any unclear areas and note questions for clarification.	All members (13/10)

2	Group Name Discussion	• Hui Yi suggested “Heng Ong Huat.” • Kang Huey proposed “404 Not Found.” • Alice Lee recommended “Server Error.” • The name “404 Not Found” was chosen by the group through a vote.	All members (13/10)
3	Software Evaluation	• Kang Huey found that “Visual Paradigm” is a powerful tool for designing floor plans, offering collaboration options for real-time editing, which allows us to create the floor plan simultaneously and efficiently. • Alice Lee discovered that “Draw.io” is an easy-to-use and free tool for creating diagrams, making it accessible for everyone to design and edit floor plans without any cost. • Hui Yi learned that “SmartDraw” is a user-friendly software with a	All members (13/10)

		wide range of templates, making it great for creating different types of diagrams. • After comparison, we decided that Visual Paradigm is the best tool for our needs.	
4	Floor Design	• Plan the layout to ensure a logical flow between different areas. • Hui Yi suggested adding a lecturer hall to accommodate larger classes and presentations. • Alice Lee recommended including a lift for accessibility between floors. • Kang Huey proposed adding toilets on both floors for convenience. • Review the layout once more to ensure it effectively accommodates the needs of all users.	All members (13/10)
5	Budget Considerations	• Discuss the budget for the project, including expenses	All members (13/10)

		for software, materials, and design. • Set a budget limit for each area to ensure effective financial management throughout the project.	
6	End Meeting	• Highlight each member's tasks to ensure everyone knows what to focus on next. • Thank everyone for their participation and concluding the meeting.	All members (13/10)

2. Second Meeting Minutes

DATE/TIME	23 Oct 2024 9am		
LOCATION	Google Meet		
AGENDA	Interview Questions and Project Feasibility Discussion		
Meeting MC	Alice Lee Hui Mee, Cheng Kang Huey, Loh Hui Yi		
ATTENDANCE			
NAME	TIME		REASON FOR ABSENCE
Alice Lee Hui Mee	0900-1030		-
Cheng Kang Huey	0900-1030		-
Loh Hui Yi	0900-1030		-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/ SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Introduction and review of task 2 requirements	- Hui Yi presents the rubric of task 2. - All members review the requirements needed for the task	All members (23/10)
2	Proposal of possible interview questions	- Alice Lee suggested four questions about security measures in each lab. - Kang Huey suggested three questions on the computer specs	All members (23/10)

		needed for each lab. • Hui Yi suggested five questions on the infrastructure and maintenance of the labs.	
3	Discuss and decide on suitable questions	• All members agree to prepare 10 questions for the interview. • All members discussed and selected the most suitable questions.	All members (23/10)
4	Distribution of tasks among members	• Each member agreed to focus on 3-4 questions and answers. • Each member will be responsible for one feasibility.	All members (23/10)
5	End meeting	• All tasks were reviewed again by all members before ending the meeting. • Ending the session after thanking all members for their contributions.	All members (23/10)

3. Third Meeting Minutes

DATE/TIME	14 Nov 2024 / 10 am		
LOCATION	Google Meet		
AGENDA	Select Devices and Task Distribution		
Meeting MC	Alice Lee Hui Mee, Cheng Kang Huey, Loh Hui Yi		
ATTENDANCE			
NAME	TIME		REASON FOR ABSENCE
Alice Lee Hui Mee	10:00-11:30		-
Cheng Kang Huey	10:00-11:30		-
Loh Hui Yi	10:00-11:30		-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/ SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Introduction and review the requirements for task 3	- Hui Yi presents the rubric of task 3. - All members review the requirements of task 3.	All members (14/11)
2	Suggestions for the LAN	Hui Yi suggested a server, firewall,	All members (14/11)

	devices needed	modem and printer as the required devices. • Kang Huey recommended a switch, PC, hub and cable. • Alice proposed a router, wireless access point (WAP) and projector	
3	Select the LAN devices for FC	• All members agree to select nine of the suggested LAN devices as the required devices for FC which are switch, router, server, cable, WAP, firewall, PC, projector and printer.	All members (14/11)
4	Discuss the reflection questions	• All members discuss the reflection questions and give the possible answers.	All members (14/11)
5	Distribution of the task among	• Each member will be responsible for	All members (14/11)

	members	three devices and one question of reflection.	
6	End meeting	Ending the meeting and thank each member for their contribution and joining the meeting.	All members (14/11)

4. Fourth Meeting Minutes

DATE/TIME	22 December 2024 / 09:50 am		
LOCATION	BK2-N28a		
AGENDA	Discussion on Device Placement and Network Connectivity		
Meeting MC	Alice Lee Hui Mee, Cheng Kang Huey, Loh Hui Yi		
ATTENDANCE			
NAME	TIME		REASON FOR ABSENCE
Alice Lee Hui Mee	09:50-11:00		-
Cheng Kang Huey	09:50-11:00		-
Loh Hui Yi	09:50-11:00		-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/ SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Introduction and review the requirements for task 4	All members reviewed the rubric and clarified the task objectives.	All members (22/12/2024)
2	Identify connections and cables	All members discussed necessary connections, agreeing on switch	All members (22/12/2024)

		ports and patch cords for proper device interconnection.	
3	Identify cable length and types	Alice recommends using Cat6 cable for connections and fiber optics as the backbone. Cable lengths are estimated based on floor plan. Hui Yi agreed, noting that this would ensure future scalability.	All members (22/12/2024)
4	Choices of cable types and connections	Kang Huey highlighted that fibre and Cat6 cables are the most suitable options for the different network segments. Alice confirmed the choices were appropriate for future growth.	All members (22/12/2024)
5	Sketch of PC and Network device	Hui Yi draws a detailed sketch of the network layout	

	arrangement	with equipment and cable routes clearly labelled. Alice reviews the sketch and confirms that it meets all the requirements.	
6	Distribution of the task among members	• Kang Huey will be responsible for designing the work areas on the floor plan and creating the 4 network diagrams for the cafe, Cisco network lab, embedded lab, and maintenance areas. • Hui Yi will handle the 4 network diagrams for the video conferencing room, hybrid classroom, rest area, and server room, as well as the cables and connections. • Alice will focus on the 4 network	All members (22/12/2024)

		diagrams for the lecturer hall, general lab, student lounge, and office areas, and will also be in charge of identifying cable length and type and preparing the meeting minutes.	
7	End meeting	The meeting closed with thanks to each member for their contribution and participation.	All members (22/12/2024)

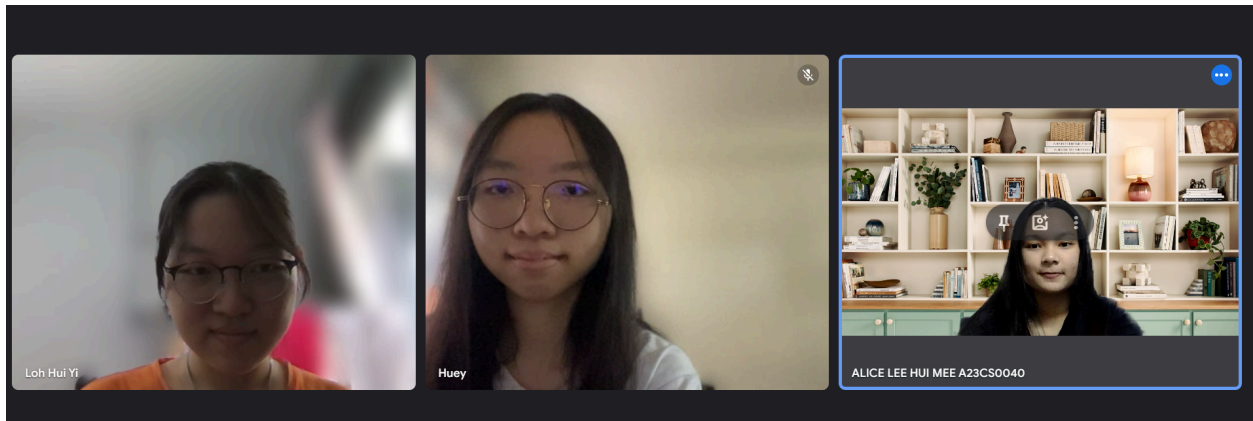
5. Fifth Meeting Minutes

DATE/TIME	30 DEC 2024 / 11 am		
LOCATION	Google Meet		
AGENDA	Understand the details about task 5 and Task Distribution		
Meeting MC	Alice Lee Hui Mee, Cheng Kang Huey, Loh Hui Yi		
ATTENDANCE			
NAME	TIME		REASON FOR ABSENCE
Alice Lee Hui Mee	11:00-12:00		-
Cheng Kang Huey	11:00-12:00		-
Loh Hui Yi	11:00-12:00		-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/ SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Introduction and review the requirements for task 5	- Alice presents the rubric of task 5. - All members review the requirements of task 5.	All members (30/12)
2	Discussion for the subnet	All members suggest and decide	All members (30/12)

		the subnet division that best suits our design. • All members try to calculate the subnet mask for each work area.	
3	IP addressing schema overview	• All members consider how the IP address is assigned to each work area.	All members (30/12)
4	Task distribution	• Alice will be in charge of the final calculation for the subnet mask and of assigning the subnet's IP address, network and broadcast addresses. • Kang Huey will be responsible for assigning the range of IP addresses for different user types based on the work area and also labeling the devices in the ground floor	All members (30/12)

		plan. Hui Yi will be in charge of labeling the devices in the first floor plan and preparing the meeting minutes.	
5	Meeting ended	The meeting ended after all the discussion was done.	All members (30/12)

6. Pictures



7. Financial Budget

No.	Details	Quantity	Amount (RM)
Income			
1.	Sponsor from Universiti Teknologi Malaysia		3,000,000.00
Expenses			
1.	Ubiquiti USW-Pro-48-PoE	9	52,011.00
2.	Archer BE230 BE3600 Dual-Band	10	4,690.00
3.	Cisco UCS C240 M5	10	13,299.10
4.	Belden USA Original Cat6 UTP Network Cable	8	4,696.00
5.	TP-Link EAP653 AX3000 Ceiling Mount WiFi 6 Access Point	4	2,304.00
6.	ASA5550-BUN-K9	8	16,634.00
7.	Acer Aspire XC-1785-14700W11S	160	703,840.00
8.	Epson EB-X51 XGA 3800 Lumens 3LCD Projector	6	12,354.00
9.	HP LaserJet Enterprise MFP M430f	4	16,052.00
10.	Labour cost	1	720,000.00
Total Expenses			1546153.10
Balance (Income - Total Expenses)			1453846.90